Module 07: Communication in Robotics — Autonomous Multi-Agent Systems

Learning Objectives

1. Define **autonomous multi-agent communication** as the exchange of beliefs, plans, and observations between self-governing robotic agents.
2. Analyze how **consensus algorithms, task allocation, and distributed planning** implement coupled Active Inference across autonomous agent teams.
3. Apply multi-agent communication frameworks to real autonomous system coordination challenges.

Introduction

Autonomous robots rarely operate alone. Self-driving cars share the road with each other, warehouse robots coordinate pickups and deliveries, drone swarms survey disaster areas, and surgical teams may include multiple robotic assistants. Communication between autonomous agents enables coordination that no individual agent could achieve alone.

Key Concepts

1. Belief Sharing and Consensus

Autonomous agents can share their probabilistic beliefs to improve collective state estimation:

- **Decentralized data fusion**: Each agent shares its local state estimate (posterior belief) with neighbors. Agents merge received beliefs with their own through precision-weighted averaging — the result is a consensus belief that incorporates evidence from multiple vantage points.
- **Consensus algorithms**: Iterative protocols where agents exchange and average estimates until the network converges on a shared belief. Convergence speed depends on network topology (fully connected: fast; chain: slow).

- **Communication bandwidth constraints**: Sharing full probability distributions is expensive. Agents often communicate compressed representations — sufficient statistics, innovation sequences, or binary flags (detection/no detection).

2. Task Allocation: Who Does What?

Multi-robot task allocation distributes work across the team:

- **Auction-based allocation**: Agents bid on tasks based on their estimated cost (Expected Free Energy of executing the task). Tasks are assigned to the lowest bidder. This is decentralized policy selection where the "team policy" minimizes total EFE across all agents.
- **Market-based approaches**: Agents trade tasks dynamically — if a robot discovers it can't complete a task efficiently (high EFE), it offers the task back to the market for reallocation.
- **Implicit allocation through stigmergy**: Agents choose tasks based on local environmental signals (e.g., a cleaning robot moves to the dirtiest area) — no explicit communication needed.

3. Distributed Planning

When tasks require coordination (two robots must simultaneously lift an object, multiple drones must maintain formation), agents must plan jointly:

- **Centralized planning with distributed execution**: A central planner computes the joint plan and distributes individual assignments. Simple but creates a single point of failure.
- **Distributed planning with communication**: Each agent plans locally but shares its planned trajectory with neighbors. Agents adjust their plans to avoid conflicts — iterating until a collision-free, coordinated plan emerges.
- **Reactive coordination protocols**: Simple rules (maintain minimum distance, match formation velocity, yield at intersections) achieve coordination without explicit planning — local Active Inference with implicit social norms.

4. Vehicle-to-Vehicle (V2V) Communication

Autonomous vehicles exemplify the need for inter-agent communication:

- V2V messages share ego-vehicle state (position, velocity, planned trajectory), sensor observations (detected obstacles, road conditions), and intent (lane change, merge, stop)
- Receiving vehicles incorporate V2V messages into their generative models as additional observation channels — precision-weighted by trust in the sending vehicle
- **Cooperative perception**: Vehicles share their sensor data to extend each agent's perceptual range beyond its own line of sight — seeing around corners through a neighbor's camera

Applications

- **Multi-drone search and rescue**: A swarm of drones searches a disaster area using auction-based task allocation — each drone bids on grid sectors based on its fuel state, distance, and sensor capability. Drones share detected survivor locations through belief-sharing protocols, enabling rapid convergence of rescue resources.
- **Autonomous intersection management**: Self-driving cars approaching an intersection without traffic lights negotiate passage through V2V communication — sharing arrival times, planned trajectories, and yielding intentions. The intersection functions as a coupled Active Inference system where vehicles jointly minimize collision risk.

Conclusion

Communication between autonomous agents — belief sharing, task allocation, distributed planning, and V2V protocols — enables coordination that transcends individual agent capabilities. These mechanisms are all implementations of coupled Active Inference, where each agent's decisions are informed by observations from other agents. The next module examines autonomous planning over extended horizons.